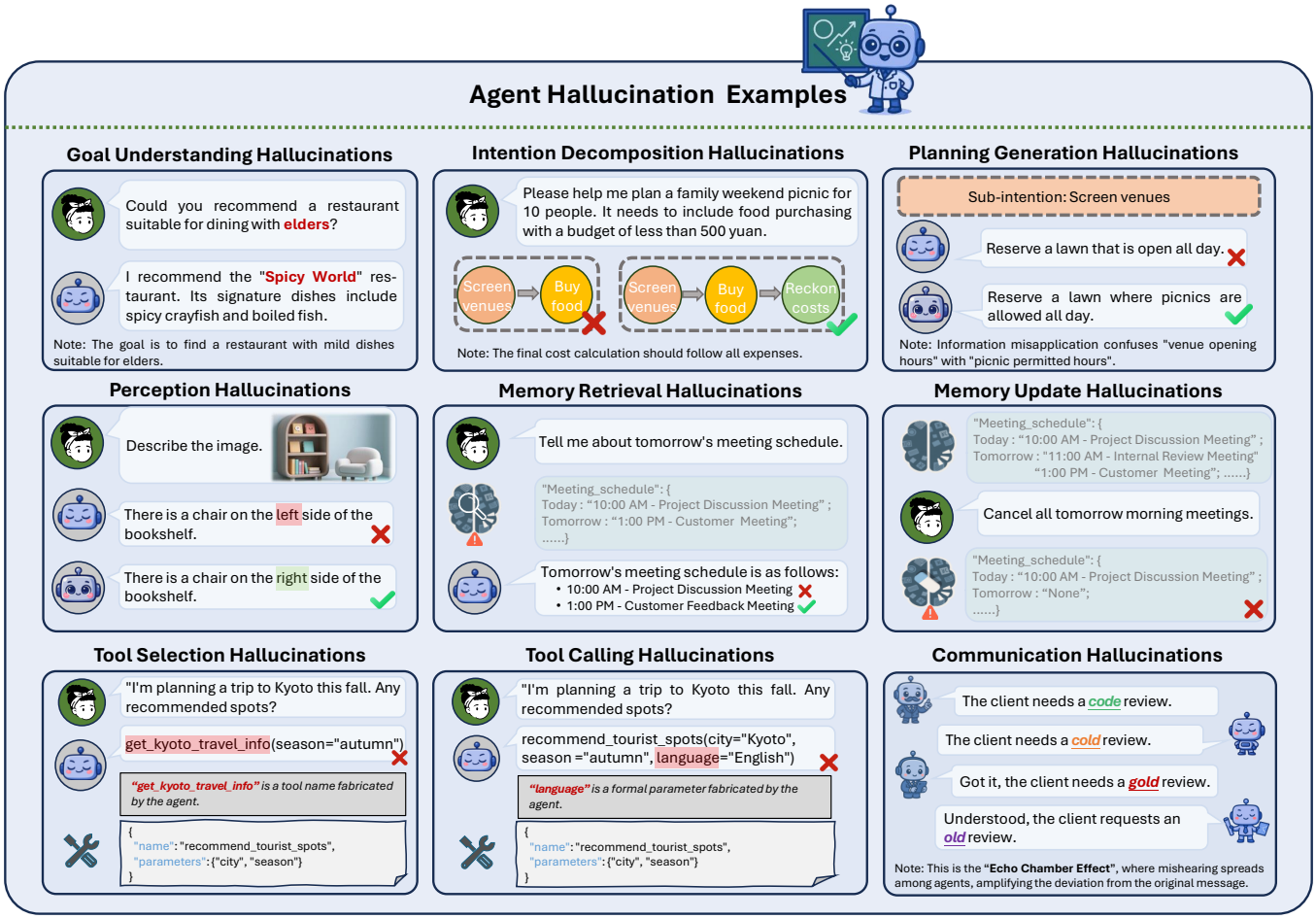

Fig. 5. The depiction of different types of agent hallucinations, each illustrated with a representative example. The detailed explanation is given in Appendix B.

agent performs an overgeneralized or incorrect update, leading to a memory update hallucination.

- **Tool Selection Hallucinations.** The user plans to travel to Kyoto in autumn and asks the agent to recommend tourist attractions. The agent incorrectly calls a non-existent tool named "get kyoto travel info", whereas the correct tool in the system should be "recommend tourist spots". This indicates that, when selecting external tools, the agent may "invent" plausible-sounding but non-existent APIs or function names, thereby exhibiting tool selection hallucinations.
- **Tool Calling Hallucinations.** Similar to the previous case, this time the agent selects the correct tool, "recommend tourist spots", but arbitrarily appends an extra parameter language="English", which is not part of the tool's actual specification. This demonstrates that even when the tool name is correct, the agent may hallucinate at the parameter level—engaging in hallucinatory argument extension—which can cause the call to fail or yield unintended behavior.
- **Communication Hallucinations.** This example illustrates an "echo chamber effect" in multi-agent communication, a particularly typical form of communication hallucination. Initially, the client states that they require a "code review". As the message passes along a chain of agents, the first agent mishears it as "cold review", the second relays it as "gold review", and the third finally interprets it as "old review". Through multiple rounds of transmission, the information is progressively distorted until it becomes completely detached from the original meaning. This hallucination shows that, when multiple agents collaborate, minor mis-hearings or mis-interpretations can be amplified, producing a "telephone game" style accumulation of deviations.